CAPSTONE PROJECT PROPOSAL SPRING 2026

Cryptocurrency Price Prediction Using LSTM and News Sentiment Analysis.

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in partial fulfillment of the requirements for the degree of
Bachelor of Science in Computer Science and Engineering

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST)

DECLARATION

We hereby declare that the work presented in this Capstone Project Proposal is the result of our own investigation and efforts, conducted under the guidance and supervision of Khandakar Md. Shafin, Lecturer, Department of Computer Science and Engineering, Bangladesh Army International University of Science and Technology (BAIUST), Cumilla, Bangladesh. We confirm that no part of this proposal has been submitted, nor is being submitted, for the award of any other degree or diploma at any institution.

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APPROVAL

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TABLE OF CONTENTS

Title		Page No.
Project/Research Title : Cryptocurrency Price Prediction Using LSTM and News Sentiment Analysis		1
Introduction		1
Project/Research Motivation		1
Problem Statement		2
REFERENCES		17

Table of Contents:

1. Introduction 1

2. Project/Research Motivation 1

3. Problem Statement 2

4. Project/Research Objectives 3

5. Related Work / Literature Review 4

6. Research Questions 5

7. Methodology / Design Approach .....(5 - 12)

7.1 Research-Based Experimental Design 5

7.2 System Features 6

7.3 Functional Requirements 6

7.4 Non-Functional Requirements 7

7.5 System Architecture 7

7.6 System Design Process 8

7.7 Professional Engineering Practices 8

7.8 Modern Tools and Technologies 8

7.9 Testing and Quality Assurance 9

7.10 Deployment and Implementation Plan 9

7.11 Research /Model / Experimental Design *(block diagram)* 10

8. Project Timeline (Gantt Chart) …… 13

9. Modern Tools & Technologies …….. 14

10. Ethical Considerations ……..……(15 - 16)

10.1 Key Ethical Considerations 15

10.2 Demonstration of Ethical Awareness 16

10.3 Commitment to Responsible Research Practices 16

11. Conclusion 16

References 17

Abbreviation

Acronym	Full form / Meaning
LTSM	Long Short Term Memory
OHCLV	Open, High, Low, Close, and Volume
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
NLP	Natural Language Processing
SMAPE	Symmetric Mean Absolute Percentage Error
CER	Certainty Equivalent Return
VADER	Valence Aware Dictionary and Sentiment Reasoners
FRED	Federal Reserve Economic Data
SHAP	Shapley Additive Explanation
API	Application Programming Interface
DSTAT	A linux command line tool
BTC	Bitcoin
ETH	Etherium
NLTK	Natural Language Toolkit

1. Introduction:

Cryptocurrency is digital money that isn't controlled by any bank or government. It exists only online and uses cryptography (secure coding) to keep transactions safe. This means no single company or authority decides how it's managed, giving users more control over their own money compared to traditional banking systems.

Instead of a bank keeping track of who owns what, a public, shared record called a blockchain does that job, verified by many computers around the world. This network of computers works together to confirm transactions, making it very difficult to fake or alter records once they're added. Bitcoin was the first and most famous example, introduced in 2009, but there are thousands of others now, such as Ethereum and Litecoin, each designed for different purposes like payments, smart contracts, or app development.

People use crypto to send money across borders quickly, invest for potential profit, or build apps like games and decentralized platforms that don't rely on a central company. However, its value can rise and fall a lot within a short time, making it a risky investment for beginners. Also, if you lose access to your account or forget your private key, there's usually no way to recover it, unlike a bank account with password reset options, so users must be extra careful in managing their crypto wallets.

2. Research Motivation

1. **Volatility problem** – Crypto currency prices change rapidly and unpredictably, making them harder to forecast than traditional assets like stocks. This creates a need for smarter prediction methods.
2. **Limits of traditional models** – Older prediction methods (like linear regression) can't handle the complex, non-linear ups and downs in crypto prices. LSTM, a deep learning model, is better suited for this kind of time-series data.
3. **Impact of news and sentiment** – Crypto prices often react strongly to news, tweets, or public opinion. Ignoring this sentiment data means missing a big part of what actually drives price changes.
4. **Practical value** – A more reliable prediction model can help investors and traders make smarter decisions in a market known for sudden crashes and spikes.
5. **Academic relevance** – As a CSE student, this project lets us apply real skills in machine learning, deep learning (LSTM), and natural language processing (sentiment analysis) to a current, real-world problem.

3. Problem Statement

Specific Issues:

- Sentiment analysis relies only on low-frequency, passive proxies (weekly Google Trends) and missing real-time social media and news sentiment
- Macroeconomic and geopolitical factors (interest rates, inflation, gold prices) are excluded from blockchain-fundamentals-based models
- Prediction models use structured price/volume (OHLCV) data only, ignoring unstructured text-based sentiment signals
- No cross-asset contagion or systemic risk modeling exists, as cryptocurrencies are treated independently
- Frequency-decomposition methods oversimplify price dynamics into only two (high/low) groups
- networks are validated on a single cryptocurrency (ADA), limiting generalizability

Solution Statement:

The proposed framework collects real-time sentiment data from social media (X/Twitter, Reddit) and crypto news portals via their APIs, and applies AI/NLP-based sentiment analysis to generate a high-frequency sentiment score. This sentiment score, combined with historical price data and macro-financial factors, will feed into a machine learning-based model to predict cryptocurrency price movements improving accuracy and responsiveness over existing low-frequency, isolated-data approaches.

4. Research Objectives

- To collect and preprocess historical cryptocurrency price data for training.
- To build an LSTM model that predicts future cryptocurrency prices based on past price patterns.
- To train and test the LSTM model using historical data to check its prediction accuracy
- To evaluate the model's performance using metrics like RMSE (Root Mean Square Error) or MAE (Mean Absolute Error).
- To compare the LSTM model's results with simpler methods (like moving averages) to see if it performs better.

Primary Objective:

The primary objective of this research is to predict cryptocurrency prices accurately using an LSTM (Long Short-Term Memory) model. This involves designing and building an LSTM model suited for time-series price prediction, and training the model using historical price data to learn underlying patterns and trends.

Secondary Objectives:

To support the primary objective, this research also involves collecting and preprocessing historical cryptocurrency price data for training and testing purposes.

The model's prediction accuracy is then evaluated using metrics such as RMSE (Root Mean Square Error) or MAE (Mean Absolute Error), and its performance is compared with simpler prediction methods like moving averages to demonstrate its effectiveness.

5. Literature Review

Cryptocurrency price prediction has become an active research area, with studies focusing on blockchain fundamentals, investor sentiment, deep learning, and decomposition-based forecasting. However, these approaches remain fragmented and often address only specific aspects of the prediction problem.

Serial	Study	Key contribution	Limitation
1	Guidolin & Ionta (2026)	Blockchain fundamentals and Google Trends sentiment across 40 cryptocurrencies.	Weekly sentiment only — no real-time social/news data.
2	Luo & Yin - CryptoForecastCF	Explainable deep learning (N-BEATS, WaveNet, Seq2Seq, GRU).	Uses OHLCV only; excludes sentiment & cross-asset links
3	Zhang et al. (2026)	Sliding EMD framework with macro variables for ADA prediction.	Single asset; only two frequency groups
4	Hajek et al. (2023)	Investor sentiment + ensemble ML improves Bitcoin prediction.	Limited to Bitcoin; simpler sentiment sources
5	Amirshahi & Lahmiri (2025)	CNN-LSTM captures nonlinear market behavior.	No integrated sentiment analysis

Other studies have further emphasized the importance of macroeconomic and market-wide factors. Corbet et al. (2020) found that cryptocurrency markets respond to monetary policy changes, while Aharon et al. (2021) and Bouri et al. (2021) reported strong relationships between exchange rates, market contagion, and cryptocurrency returns. Hajek et al. (2023) also showed that combining investor sentiment with ensemble machine learning improves Bitcoin price prediction.

Despite these advances, no existing study integrates blockchain fundamentals, high-frequency multi-source sentiment, macroeconomic indicators, and advanced forecasting techniques within a single unified framework. This research aims to address this gap by combining AI/NLP-based sentiment analysis with blockchain and macro-financial variables to improve cryptocurrency price prediction and portfolio performance.

Research Gap:
Existing studies address sentiment, blockchain fundamentals, interpretability, or frequency decomposition largely in isolation. Guidolin and Ionta (2026) rely on low-frequency sentiment and omit macroeconomic factors; Luo and Yin's framework ignores unstructured sentiment and contagion effects; and Zhang et al. (2026) apply only a two-group frequency split validated on a single asset. This project addresses these gaps by integrating real-time, AI/NLP-derived sentiment (from social media and news APIs) with macro-financial variables within a machine learning-based prediction framework, aiming to improve both forecasting accuracy and generalizability across cryptocurrencies.

6. Research Questions:

RQ1. How accurately can an LSTM (Long Short-Term Memory) model predict cryptocurrency prices using historical price data?

RQ2. Does incorporating news sentiment analysis improve the accuracy of cryptocurrency price predictions compared to using historical price data alone?

RQ3. What is the relationship between public sentiment (from news/social media) and cryptocurrency price movements?

RQ4. How does a combined model (LSTM + sentiment analysis) perform compared to a model using only price data, in terms of accuracy and reliability?

7. Development Methodology

7.1 Research-Based Experimental Design with Iterative Refinement.

Phase	Description
1. Literature Review & Requirement Gathering	Reviewed existing research (Guidolin & Ionta, 2026; Luo & Yin; Zhang et al., 2026) to identify gaps in sentiment frequency, factor integration, and decomposition methods.
2. Data Collection & System Design	Collected price/volume (OHLCV) data, social media & news sentiment (via APIs), and macro-financial data; designed the data pipeline and model architecture blueprint.
3. Model Development & Implementation	Developed sentiment analysis (NLP-based) and price prediction models (ML/Deep Learning) using Python, TensorFlow/Keras, and Scikit-learn.
4. Testing & Validation	Performed out-of-sample, walk-forward validation; evaluated using RMSE, MAE, SMAPE, Sharpe ratio, and CER metrics.
5. Deployment / Result Integration	Integrated final model outputs into a dashboard/visualization for prediction results and sentiment trends.
6. Refinement & Optimization	Hyperparameter tuning, model comparison, and error analysis to improve prediction accuracy.
7. Documentation	Technical manuals, user guides, and project evaluation reports.

7.2 System Features

Data Collection Module

- Real-time price/volume (OHLCV) data collection from crypto exchanges (Binance, CoinMarketCap)
- Social media sentiment data collection via X/Twitter and Reddit APIs
- News sentiment data collection via crypto news portal APIs
- Macroeconomic data collection (interest rates, inflation, gold prices) from sources like FRED

Sentiment Analysis Module

- NLP/AI-based sentiment scoring of collected text data
- Daily/intraday sentiment index generation
- Sentiment trend visualization over time

Prediction Module

- Machine learning/deep learning-based price movement prediction
- Integration of sentiment scores with price and macro-financial features
- Model retraining and rolling-window forecasting

Analytics & Reporting Module

- Prediction accuracy reports (RMSE, MAE, SMAPE)
- Comparison dashboard: proposed model vs. baseline (weekly sentiment) model
- Historical performance tracking (Sharpe ratio, CER)

Visualization/Dashboard Module

- Real-time sentiment and price trend charts
- Prediction result display (predicted vs. actual price movement)
- Alerts for significant sentiment shifts or predicted volatility

7.3 Functional Requirements

ID	Requirement	Description
FR1	Data Ingestion	Automatically collect price/volume (OHLCV), sentiment (social media/news), and macroeconomic data via APIs.
FR2	Sentiment Scoring	Apply NLP-based sentiment analysis to generate daily/intraday sentiment scores from collected text data.
FR3	Data Preprocessing	Clean, normalize, and align sentiment, price, and macro data on a consistent time frequency.
FR4	Price Prediction	Generate machine learning/deep learning-based predictions of cryptocurrency price movement or direction.
FR5	Model Evaluation	Compute and display prediction accuracy metrics (RMSE, MAE, SMAPE, DSTAT).
FR6	Baseline Comparison	Compare proposed model's performance against baseline (e.g., weekly Google Trends-only) models.
FR7	Reporting/Visualization	Generate charts and reports showing sentiment

		trends, predicted vs. actual price movement.
FR8	Alert/Notification	Notify significant sentiment shifts or high-confidence prediction signals.

7.4 Non-Functional Requirements

Category	Requirement
Performance	Process daily/intraday data updates with prediction latency under a few seconds per cryptocurrency.
Security	Secure API key storage and encrypted data handling for third-party API access (social media/news/exchange APIs).
Usability	Clear, interpretable dashboard/visualization for sentiment trends and prediction results.
Scalability	Ability to extend the framework to additional cryptocurrencies without major architectural change.
Reliability	Consistent data pipeline uptime to avoid gaps in sentiment/price data collection.
Compatibility	Compatible with standard data science environments (Python-based, cloud or local execution).

7.5 System Architecture

Layers:

1. **Data Collection Layer:** Yahoo Finance, Binance API (price/volume data); CryptoPanic API or NewsAPI (crypto news sentiment); social media APIs (X/Twitter, Reddit) – gathers raw price and sentiment input data.
2. **Processing Layer:** Python, Pandas, NumPy – Data cleaning, normalization, and feature engineering.
3. **Sentiment Analysis Layer:** NLTK, VADER, Transformers – NLP-based sentiment scoring of collected text data.
4. **Modeling Layer:** Scikit-learn (ML models), TensorFlow/Keras (deep learning models) – Price prediction using sentiment, price, and macro features.
5. **Visualization Layer:** Matplotlib, Plotly – Sentiment trend charts, prediction vs. actual price comparison, model performance dashboards.

7.6 System Design Process

- **Data Flow Diagram (DFD):** Illustrates the flow from data collection (Yahoo Finance/Binance API, CryptoPanic/NewsAPI) → preprocessing (Pandas/NumPy) → sentiment scoring (NLTK/VADER/Transformers) → model prediction (Scikit-learn/TensorFlow) → visualization (Matplotlib/Plotly).
- **Model Architecture Diagram:** Depicts input features (sentiment scores, price/volume, macro variables), model layers, and output (predicted price/direction).
- **Design Principles:** Modularity (separate sentiment, price, and macro pipelines), reproducibility (fixed random seeds, version-controlled notebooks), scalability (extendable to additional cryptocurrencies).

- **Development Environment:** Jupyter Notebook for prototyping and iterative experimentation; GitHub for version control.

7.7 Professional Engineering Practices

- **Git/GitHub version control** — track code changes, model versions, and experiment history
- **Reproducible research practices** — fixed random seeds, documented preprocessing steps, saved model checkpoints
- **Peer/supervisor review** — periodic review of methodology, code, and results for correctness
- **Ethical and compliant API usage** — respecting rate limits and terms of service of social media, news, and exchange APIs; no scraping of restricted/private data
- **Standard coding practices** — clean, modular Python code (PEP8 style), clear inline documentation and comments
- **Data integrity practices** — avoiding data leakage (e.g., using only past information in rolling/walk-forward validation)

7.8 Modern Tools and Technologies

Category	Tool/Technology	Purpose
Programming	Python	Core language for data processing and modeling
Data Analysis	Pandas, NumPy	Data cleaning, transformation, feature engineering
Visualization	Matplotlib, Plotly	Sentiment trend charts, prediction result plots
Machine Learning	Scikit-learn	Traditional ML models (RF, SVR, XGBoost)
Deep Learning	TensorFlow/Keras	LSTM/GRU-based sequential prediction models
NLP/Sentiment Analysis	NLTK, VADER, Transformers	Sentiment scoring from text (social media/news)
Price Data Source	Yahoo Finance, Binance API	Historical and real-time cryptocurrency price data
News/Sentiment Source	CryptoPanic API, NewsAPI	Crypto-related news for sentiment extraction
Version Control	Git, GitHub	Code and experiment version tracking
Development Environment	Jupyter Notebook	Interactive prototyping and analysis

7.9 Testing and Quality Assurance

Test Type	Purpose	Test Type
Data Validation Testing	Verify completeness and correctness of collected price/sentiment/macro data	Pandas assertions, manual checks
Unit Testing	Validate individual functions (data cleaning, sentiment scoring, feature engineering)	Python unittest/pytest
Model Validation Testing	Ensure model trains and predicts without errors on sample data	Scikit-learn/TensorFlow validation splits
Out-of-Sample Testing	Evaluate model performance on unseen, chronologically later data	Walk-forward/rolling window validation
Performance Testing	Assess prediction accuracy and computational efficiency	RMSE, MAE, SMAPE, runtime benchmarking
Robustness Testing	Confirm model generalizes across multiple cryptocurrencies	Cross-asset testing (BTC, ETH, ADA, etc.)

7.10 Deployment and Implementation Plan

- **Hosting/Execution:** Local or cloud-based execution (Google Colab / Jupyter Notebook); no production-grade server hosting required, as this is a research/model validation project rather than a live user-facing application
- **Environment Management:** Python virtual environment (venv/conda) with version-pinned dependencies (requirements.txt) for reproducibility
- **Data Backup:** Periodic backup of collected datasets (price, sentiment, macro data) and trained model checkpoints to cloud storage (Google Drive/GitHub)
- **Release Plan:** Prototype (single-cryptocurrency model) → Extended validation (multi-cryptocurrency testing) → Final research report/thesis submission with reproducible code repository
- **7.11 Experimental Design** (block diagram)

This research adopts a data-driven, experimental design combining sentiment data collection, macro-financial factor integration, and machine learning-based model development to predict cryptocurrency price movements. The study uses quantitative methods — predictive

modeling, out-of-sample forecasting, and statistical performance evaluation — to validate the proposed framework against existing sentiment-and-fundamentals-based approaches.

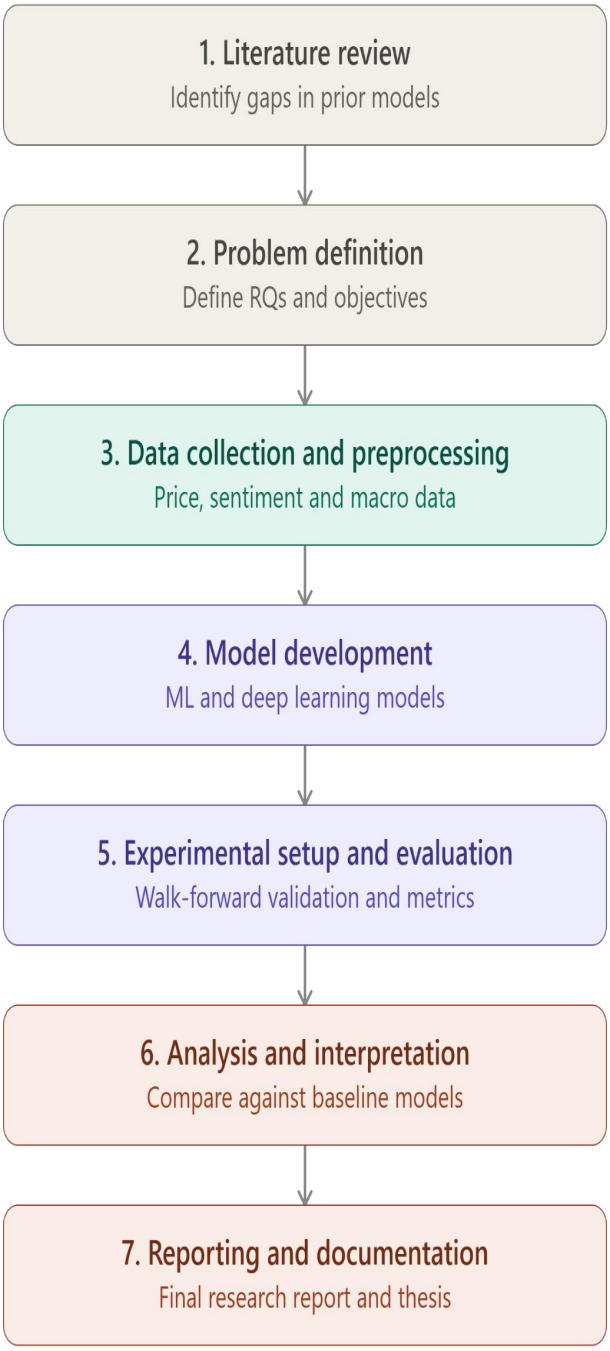


Fig (1) : Experimental Design Block Diagram

Methodology Steps in Details

1. **Problem Definition & Research Questions**

- Define the core research problem: the lag and incompleteness of existing sentiment-and-fundamentals-based crypto prediction models
- Formulate research questions guiding the design of the sentiment collection pipeline, factor integration, and predictive model
- Ensure alignment between the identified gaps (weekly-only sentiment, missing macro factors, single-asset validation) and the proposed ML-based solution

2. **Data Collection & Preprocessing**

Data Sources: Cryptocurrency price/volume (OHLCV) data via Yahoo Finance and Binance API; sentiment data from social media (X/Twitter, Reddit) and crypto news portals via CryptoPanic API or NewsAPI; macroeconomic data (interest rates, inflation, gold prices) from sources such as FRED

Preprocessing Steps:

- Data cleaning: handle missing values, remove duplicate/erroneous entries
- Sentiment scoring: apply NLP-based sentiment analysis (NLTK, VADER, or Transformer-based models) to generate daily/intraday sentiment scores
- Feature selection: select relevant price, sentiment, and macro features
- Normalization/standardization: scale numeric features (Min-Max or Z-score normalization)
- Data splitting: partition into training (70%), validation (15%), and test (15%) sets, preserving chronological order to prevent data leakage

3. **Model/Methodology Development**

- Develop multiple predictive models to evaluate performance and robustness:
- **Machine Learning Models:** Random Forest, Gradient Boosting, XGBoost, Support Vector Regression (via Scikit-learn)
- **Deep Learning Models:** LSTM, GRU-based architectures for sequential sentiment-price data (via TensorFlow/Keras)
- Implement hyperparameter tuning via grid search or Bayesian optimization
- Employ walk-forward (rolling window) cross-validation to preserve temporal order and prevent look-ahead bias

4. **Experimental Setup & Evaluation**

- Implement models in Python using TensorFlow/Keras, Scikit-learn, and Pandas/NumPy
- Evaluate using metrics such as:
- RMSE, MAE, SMAPE for prediction accuracy
- Directional Accuracy (DSTAT) for trend prediction
- Sharpe ratio, CER (Certainty Equivalent Return) for investment/economic performance
- Compare performance against baseline models (e.g., weekly Google Trends-only sentiment model, non-decomposed univariate model)
- Conduct statistical significance tests (e.g., Diebold-Mariano test) to validate improvements

5. **System/Prototype Integration**

- Develop a lightweight analysis pipeline/dashboard integrating:
- Real-time sentiment collection module (API-based)
- Price and macro data ingestion module
- Prediction model generating forecasted price direction/returns
- Visualization (Matplotlib/Plotly) for sentiment trends and prediction accuracy
- Technologies: Python, Jupyter Notebook for prototyping; GitHub for version control

6. **Testing & Validation**

- Conduct out-of-sample backtesting on historical data to assess real-world applicability
- Validate model robustness across multiple cryptocurrencies (large-cap, mid-cap, PoW/PoS coins)
- Measure impact on prediction reliability: forecast accuracy improvement, reduction in sentiment lag, and economic value (CER/Sharpe ratio) compared to baseline models

7. **Analysis & Interpretation of Results**

- Analyze model performance across different cryptocurrencies and market conditions (bull/bear phases)
- Interpret results using Explainable AI (XAI) techniques such as SHAP or counterfactual explanations to identify which sentiment/macro features drive predictions
- Identify limitations (e.g., API rate limits, data availability, model overfitting) and areas for future improvement

8. **Reporting & Documentation**

- Document methodology, experimental results, and model evaluation clearly
- Prepare findings for the final research report/thesis, with potential extension toward conference or journal submission

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8. Estimated Project Timeline (Gantt Chart)

SL.	Phase	Duration (Days)	Start Day	End Day	Dependencies
1.	Literature Review	7	1	7	None
2.	Problem Definition & Research Questions	5	8	12	Literature Review
3.	Data Collection (Price/Sentiment/Macro APIs)	10	13	22	Problem Definition
4.	Sentiment Scoring & Preprocessing	10	18	27	Data Collection (parallel, starts mid-collection)
5.	Model Development (ML/DL)	15	28	42	Sentiment Scoring & Preprocessing
6.	Experimental Setup & Evaluation	10	43	52	Model Development
7.	Analysis & Interpretation	7	53	59	Experimental Setup & Evaluation
8.	Documentation & Final Report	10	55	64	Analysis (parallel, starts during evaluation)

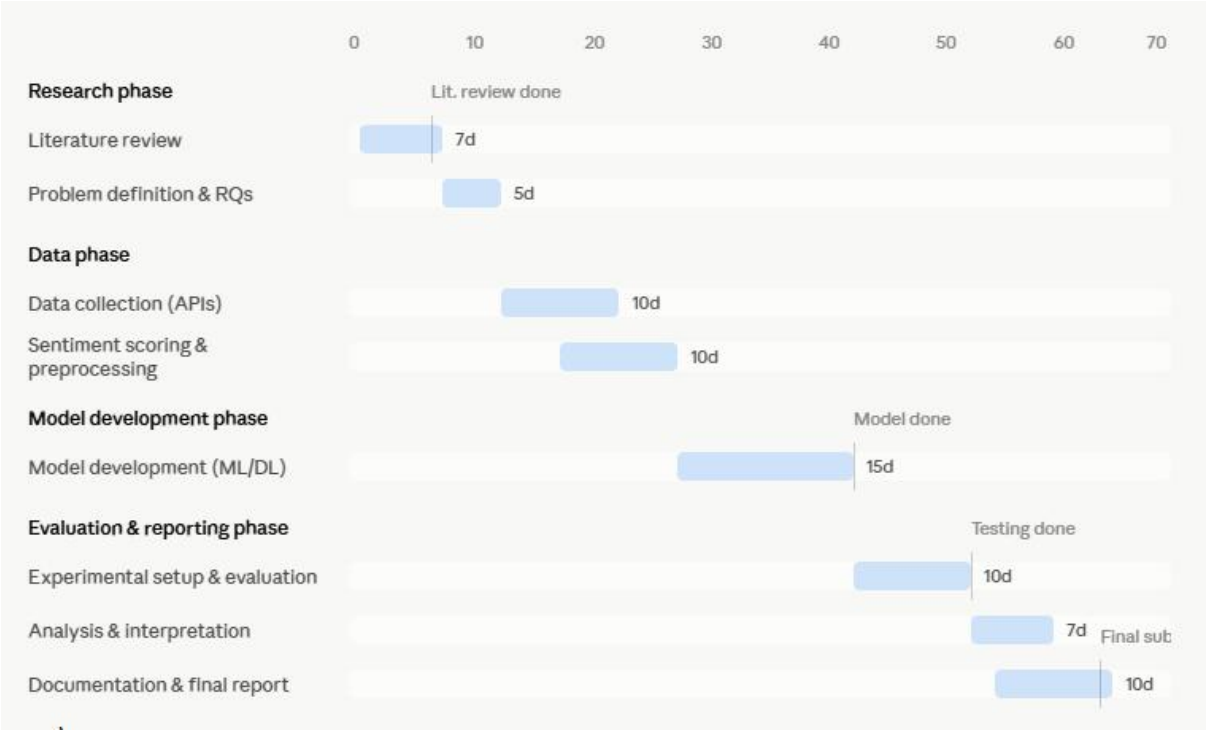


Fig2 : Project Timeline Diagram

Phase Descriptions & Milestones

1. **Literature Review (Days 1–7):** Reviewing existing research on cryptocurrency sentiment analysis, blockchain fundamentals, and decomposition-based prediction

models (Guidolin & Ionta, 2026; Luo & Yin; Zhang et al., 2026); identifying research gaps.

Milestone: Literature review completed and gaps finalized.

2. **Problem Definition & Research Questions (Days 8–12):** Defining the core research problem, objectives, and specific research questions based on identified gaps.

   Milestone: Problem statement and research questions approved.

3. **Data Collection & Sentiment Preprocessing (Days 13–27):** These are performed with overlapping schedules to maximize efficiency. Data collection focuses on gathering price/volume, sentiment, and macroeconomic data via APIs; sentiment preprocessing focuses on NLP-based scoring of collected text data.

   Milestone: Cleaned, sentiment-scored dataset ready for modeling.

4. **Model Development (Days 28–42):** Building and training machine learning/deep learning models (Random Forest, XGBoost, LSTM/GRU) that integrate sentiment, price, and macro features.

   Milestone: Trained models ready for evaluation.

5. **Experimental Setup & Evaluation (Days 43–52):** Out-of-sample, walk-forward backtesting; computing accuracy metrics (RMSE, MAE, SMAPE) and comparing against baseline models.

   Milestone: Evaluation results validated and benchmarked.

6. **Analysis & Interpretation (Days 53–59):** Analyzing model performance across cryptocurrencies, interpreting results using explainability techniques (e.g., SHAP), identifying limitations.

   Milestone: Final results interpreted and documented.

7. **Documentation & Final Report (Days 55–64):** Finalizing the research report/thesis, methodology write-up, and results summary (begins while analysis finishes).

   Milestone: Final research report submitted

9. Modern Tools & Technologies

- **Programming Languages:** Python 3.11
- **Libraries:** Scikit-learn, TensorFlow 2.x/Keras, XGBoost, Pandas, NumPy, NLTK, VADER, Transformers (Hugging Face)
- **Data Sources/APIs:** Yahoo Finance, Binance API (price data); CryptoPanic API or NewsAPI (news sentiment); X/Twitter API, Reddit API (social media sentiment); FRED (macroeconomic data)
- **Visualization Tools:** Matplotlib, Plotly

- **Development Environment:** Jupyter Notebook, VS Code
- **Version Control:** Git, GitHub
- **Testing/Validation:** Walk-forward (rolling window) cross-validation, Diebold-Mariano statistical test
- **Explainability Tools:** SHAP

10. Ethical Considerations

10.1 Key Ethical Considerations

Data Usage & API Compliance – Terms of service and authorized access:

- Collect social media (X/Twitter, Reddit), news, and price data strictly through official, authorized APIs, respecting each platform's rate limits and terms of service.
- Avoid scraping or accessing any private, restricted, or non-public data.
- Use only publicly available posts/comments for sentiment analysis; do not attempt to identify or profile individual users.

Data Privacy – Handling of user-generated content:

- Anonymize or aggregate social media/news text data used for sentiment scoring; do not store or report identifiable usernames, profiles, or personal information.
- Limit data collection to what is necessary for sentiment analysis and price prediction (text content, timestamps, aggregate scores).
- Securely store collected data with access limited to the research team.

Fairness & Bias – Avoiding skewed sentiment or model bias:

- Ensure sentiment analysis models are evaluated for bias across different sources (e.g., not overly weighted toward a single platform or a small group of highly active accounts).
- Regularly validate that sentiment scores and predictions are not systematically skewed by bot activity, spam, or coordinated posting.
- Test model performance across multiple cryptocurrencies to avoid conclusions biased toward a single asset.

Transparency & Explainability – Model clarity and limitations:

- Clearly document the sentiment scoring methodology, model architecture, features used, and evaluation metrics.
- Disclose the assumptions and limitations of the prediction model (e.g., not a guaranteed financial forecasting tool).
- Make clear that this research is for academic/analytical purposes and not intended as direct financial or investment advice.

Data Security – Safe handling of API keys and datasets:

- Store API keys and credentials securely (e.g., environment variables, not hard-coded or publicly shared).
- Maintain secure, access-controlled storage for collected datasets and trained models.
- Follow responsible disclosure if any data pipeline vulnerability or leakage risk is identified.

Accountability – Responsible reporting of results:

- Clearly report both the strengths and limitations of the proposed model, avoiding overstated claims about prediction accuracy or profitability.
- Take responsibility for correcting errors in methodology, data handling, or reported results if identified.
- Comply with academic integrity standards and institutional research ethics guidelines.

10.2 Demonstration of Ethical Awareness

This research involves collecting and analyzing publicly available social media, news, and financial market data to build a cryptocurrency sentiment and price prediction framework. Ethical considerations include respecting data source terms of service, protecting the privacy of individuals whose public posts are used for sentiment analysis, avoiding biased or misleading model claims, and being transparent about the limitations of the prediction framework — particularly given the speculative and volatile nature of cryptocurrency markets. Given the potential for financial decision-making impact, all project activities will avoid presenting results as investment advice and will emphasize the academic/research nature of the findings.

10.3 Commitment to Responsible Research Practices

We are committed to conducting this research responsibly by:

- Collecting data only through authorized, terms-compliant APIs
- Protecting the privacy of individuals in publicly sourced social media/news data
- Avoiding biased, overstated, or misleading claims about model performance
- Transparently documenting methodology, data sources, and model limitations
- Clearly stating that this research is for academic purposes and not financial advice
- Taking responsibility for the accuracy and integrity of reported results

This ensures the cryptocurrency sentiment and price prediction research is conducted transparently, responsibly, and with appropriate academic and ethical rigor.

11. Conclusion

This research proposes an AI-powered cryptocurrency price prediction framework that integrates high-frequency sentiment analysis with macro-financial factors, addressing key gaps in existing literature such as low-frequency sentiment proxies and limited

macroeconomic integration. Through API-based data collection, NLP-based sentiment scoring, and rigorous out-of-sample evaluation, the study aims to improve prediction accuracy while maintaining ethical and transparent research practices.

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